

MATH-334 Numerical Analysis

Newton's Divided Difference Interpolations

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Outline

① Topics Related to Interpolation Covered So Far

Curve Fitting, Regression and Interpolation

Why Polynomial Interpolation?

Polynomial Interpolation Problem

Lagrange Interpolating Polynomials

② Newton Divided Difference Interpolation

Derivation of Lagrange Int. for $n+1$ points

Newton Divided Difference (NDD) Table

Constructing NDD Polynomial with NDD Table

③ Summary

Polynomial Interpolation Problem

- Given**
- ① The $(n + 1)$ nodes: $\underbrace{x_0, x_1, \dots, x_n}_{n+1}$ **Data points**
 - ② The functional values

$f(x_0), f(x_1), \dots, f(x_n)$ on nodes $x_0, x_1, \dots, x_n$.

- ③ An Intermediate (non-tabulated) point: x^*

Predict $f(x^*)$: the value at $x = x^*$.

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- ⊛ Polynomial interpolation consists of determining the unique polynomial $y = P(x)$ of order n that fits $n + 1$ data points.
- ⊛ This **polynomial provides a formula** to compute intermediate values.

- Lagrange Interpolating Polynomials

x_i	2	2.5	4
$f(x_i)$	1/2	1/2.5	1/4

$$P(x) = L_0(x)f(x_0) + L_1(x)f(x_1) + L_2(x)f(x_2)$$

Step 1.

$$L_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} = \frac{(x - 2.5)(x - 4)}{(-0.5)(-2)} = (x - 2.5)(x - 4) = x^2 - 6.5x + 10$$

$$L_1(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} = \frac{(x - 2)(x - 4)}{(0.5)(-1.5)} = -\frac{1}{0.75}(x - 2)(x - 4) = \frac{x^2 - 6x + 8}{0.75}$$

$$L_2(x) = \frac{(x - x_1)(x - x_0)}{(x_2 - x_1)(x_2 - x_0)} = \frac{1}{3}(x - 2.5)(x - 2) = \frac{x^2 - 4.5x + 5}{3}$$

$$L_0(x) = x^2 - 6.5x + 10, \quad L_1(x) = \frac{x^2 - 6x + 8}{0.75} \text{ and } L_2(x) = \frac{x^2 - 4.5x + 5}{3}$$

Step 2. Find the interpolating polynomial $P_2(x)$:

$$\begin{aligned} P_2(x) &= f(x_0)L_0(x) + f(x_1)L_1(x) + f(x_2)L_2(x) = \frac{1}{2}L_0(x) + \frac{1}{2.5}L_1(x) + \frac{1}{4}L_2(x) \\ &= \frac{1}{2}(x^2 - 6.5x + 10) + \frac{1}{2.5} \frac{(x^2 - 6x + 8)}{0.75} + \frac{1}{4} \frac{(x^2 - 4.5x + 5)}{3} \end{aligned}$$

$$P_2(3) = 0.3250.$$

For the data

x_i	0	1	2	4
$f(x_i)$	7	13	21	43

find the polynomial we will use the following formula

$$P_3(x) = 7L_0(x) + 13L_1(x) + 21L_2(x) + 43L_3(x)$$

$$L_0(x) = \frac{(x-1)(x-2)(x-4)}{-8}$$

$$L_1(x) = \frac{(x)(x-2)(x-4)}{3}$$

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For $P_3(x)$ at $x = 3$

$$L_0(3) = \frac{(3-1)(3-2)(-4)}{-8} = \frac{1}{4}, \quad L_1(3) = \frac{(3)(3-2)(3-4)}{3} = -1$$

$$L_2(3) = \frac{(3)(3-1)(3-4)}{-4} = \frac{3}{2}, \quad L_3(3) = \frac{(3)(3-1)(3-2)}{24} = \frac{1}{4}$$

$$\text{Thus } P_3(3) = 7\left(\frac{1}{4}\right) + 13(-1) + 21\left(\frac{3}{2}\right) + 43\left(\frac{1}{4}\right) = 31$$

- **Newton Divided Difference Interpolation**
- *Derivation of Newton Divided Difference*

Suppose that $P(x)$ is the n^{th} Lagrange polynomial that agrees with the function f at the distinct numbers $\underbrace{x_0, x_1, \dots, x_n}_{n+1 \text{ data points}}$.

$$\begin{aligned} P(x) &= L_{n,0}(x)f(x_0) + L_{n,1}(x)f(x_1) + \dots + L_{n,n}(x)f(x_n) \\ &= \sum_{k=0}^n f(x_k)L_k(x), \quad \text{where } L_k(x) = \prod_{\substack{i=0 \\ i \neq k}}^n \frac{x - x_i}{x_k - x_i}, \end{aligned}$$

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Although this polynomial is unique, there are alternate algebraic representations that are useful in certain situations. One of these situations is **Newton's Divided Differences of f** with respect to $x_0, x_1, \dots, x_n$.

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These constants $a_0, a_1, \cdots, a_n$ must depend on given data that is

x_0	x_1	$\cdot \quad \cdots$	x_n
$f(x_0) = f_0$	$f(x_1) = f_1$	$\cdot \quad \cdots$	$f(x_n) = f_n$

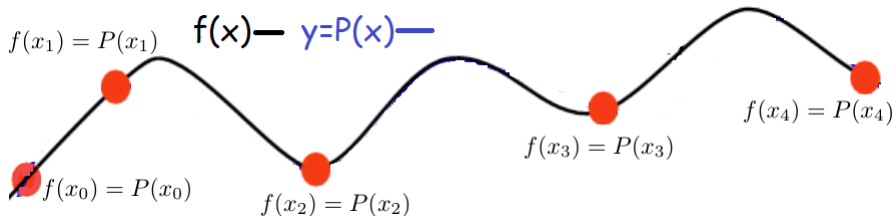
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$P(x)$ at x_2 gives

$$P(x_2) = a_0 + a_1(x_2 - x_0) + a_2(x_2 - x_0)(x_2 - x_1)$$

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Comparing a_0, a_1 and a_2 can we get some pattern?

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First order Newton's polynomial

The equation of line passing through (x_0, f_0) and (x_1, f_1) is given by

$$\begin{aligned}
 \frac{f(x) - f_1}{x - x_1} &= \frac{f_1 - f_0}{x_1 - x_0} \\
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$$\begin{array}{rcl}
 \underbrace{a_0}_{\text{depends on } x_0} & = & f_0 \\
 \underbrace{a_1}_{\text{depends on } x_0 \text{ and } x_1} & = & \frac{f_1 - f_0}{x_1 - x_0} \\
 \underbrace{a_2}_{\text{depends on } x_0 \text{ } x_1 \text{ and } x_2} & = & \frac{\frac{f_2 - f_1}{x_2 - x_1} - \frac{f_1 - f_0}{x_1 - x_0}}{x_2 - x_0}
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Comparing a_0, a_1 and a_2 can we get some pattern?

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 & & \underbrace{a_2}_{\text{depends on } x_0 \text{ } x_1 \text{ and } x_2} &= \frac{\frac{f_2 - f_1}{x_2 - x_1} - \frac{f_1 - f_0}{x_1 - x_0}}{x_2 - x_0}
 \end{aligned}$$

If we denote $a_0 = f_0$ as $f[x_0]$. Then its easy to write a_1 as

$$a_1 = \frac{f[x_1] - f[x_0]}{x_1 - x_0}$$

Comparing a_0, a_1 and a_2 can we get some pattern?

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 \underbrace{a_0}_{\text{depends on } x_0} &= f_0 \\
 \underbrace{a_1}_{\text{depends on } x_0 \text{ and } x_1} &= \frac{f_1 - f_0}{x_1 - x_0} \\
 \underbrace{a_2}_{\text{depends on } x_0, x_1 \text{ and } x_2} &= \frac{\frac{f_2 - f_1}{x_2 - x_1} - \frac{f_1 - f_0}{x_1 - x_0}}{x_2 - x_0}
 \end{aligned}$$

If we denote $a_0 = f_0$ as $f[x_0]$. Then its easy to write a_1 as

$$a_1 = \frac{f[x_1] - f[x_0]}{x_1 - x_0}$$

If we denote $a_1 = f[x_0, x_1]$. Then

$$a_2 = \frac{f[x_2, x_1] - f[x_1, x_0]}{x_2 - x_0}$$

By looking at the pattern of a_1 and a_2 is it possible to write a_3 ?

$$\underbrace{a_1}_{f[x_1, x_0]} = \frac{f[x_1] - f[x_0]}{x_1 - x_0}$$

$$\underbrace{a_2}_{f[x_2, x_1, x_0]} = \frac{f[x_2, x_1] - f[x_1, x_0]}{x_2 - x_0}$$

By looking at the pattern of a_1 and a_2 is it possible to write a_3 ?

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$$\underbrace{a_2}_{f[x_2, x_1, x_0]} = \frac{f[x_2, x_1] - f[x_1, x_0]}{x_2 - x_0}$$

$$\underbrace{a_3}_{f[x_3, x_2, x_1, x_0]} =$$

By looking at the pattern of a_1 and a_2 is it possible to write a_3 ?

$$\underbrace{a_1}_{f[x_1, x_0]} = \frac{f[x_1] - f[x_0]}{x_1 - x_0}$$

$$\underbrace{a_2}_{f[x_2, x_1, x_0]} = \frac{f[x_2, x_1] - f[x_1, x_0]}{x_2 - x_0}$$

$$\underbrace{a_3}_{f[x_3, x_2, x_1, x_0]} = \frac{\overbrace{f[x_3, x_2, x_1]}^{\text{First three}} - \overbrace{f[x_2, x_1, x_0]}^{\text{Last three}}}{x_3 - x_0}$$

By looking at the pattern of a_1 and a_2 is it possible to write a_3 ?

$$\underbrace{a_1}_{f[x_1, x_0]} = \frac{f[x_1] - f[x_0]}{x_1 - x_0}$$

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$$\underbrace{a_3}_{f[x_3, x_2, x_1, x_0]} = \frac{\overset{\text{First three}}{f[x_3, x_2, x_1]} - \overset{\text{Last three}}{f[x_2, x_1, x_0]}}{x_3 - x_0}$$

$$\underbrace{a_k}_{f[x_0, x_1, \dots, x_k]} =$$

By looking at the pattern of a_1 and a_2 is it possible to write a_3 ?

$$\underbrace{a_1}_{f[x_1, x_0]} = \frac{f[x_1] - f[x_0]}{x_1 - x_0}$$

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$$\underbrace{a_3}_{f[x_3, x_2, x_1, x_0]} = \frac{\overset{\text{First three}}{f[x_3, x_2, x_1]} - \overset{\text{Last three}}{f[x_2, x_1, x_0]}}{x_3 - x_0}$$

$$\underbrace{a_k}_{f[x_0, x_1, \dots, x_k]} = \frac{\overset{\text{First } k}{f[\underbrace{\dots}_{k \text{ points}}]} - \overset{\text{Last } k}{f[\underbrace{\dots}_{k \text{ points}}]}}{x_k - x_0}$$

Newton's Divided Difference

Hence, the interpolating polynomial (1) may be expressed as

$$P(x) = f[x_0] + f[x_1, x_0](x - x_0) + a_2(x - x_0)(x - x_1) + \cdots + a_n(x - x_0)(x - x_1) \cdots (x - x_{n-1}),$$

where

$$a_k = f[x_k, x_{k-1}, \cdots, x_1, x_0] \quad \text{for } k = 0, 1, \cdots, n.$$

$$f[x_i] = f(x_i) \quad \text{zeroth divided difference}$$

$$f[x_{i+1}, x_i] = \frac{f[x_{i+1}] - f[x_i]}{x_{i+1} - x_i} \quad 1^{st} \text{ divided difference}$$

$$f[x_{i+2}, x_{i+1}, x_i] = \frac{f[x_{i+2}, x_{i+1}] - f[x_{i+1}, x_i]}{x_{i+2} - x_i} \quad 2^{nd} \text{ divided difference}$$

$$f[x_0, x_1, \cdots, x_k] =$$

Newton's Divided Difference

Hence, the interpolating polynomial (1) may be expressed as

$$P(x) = f[x_0] + f[x_1, x_0](x - x_0) + a_2(x - x_0)(x - x_1) + \cdots + a_n(x - x_0)(x - x_1) \cdots (x - x_n),$$

where

$$a_k = f[x_k, x_{k-1}, \cdots, x_1, x_0] \quad \text{for } k = 0, 1, \cdots, n.$$

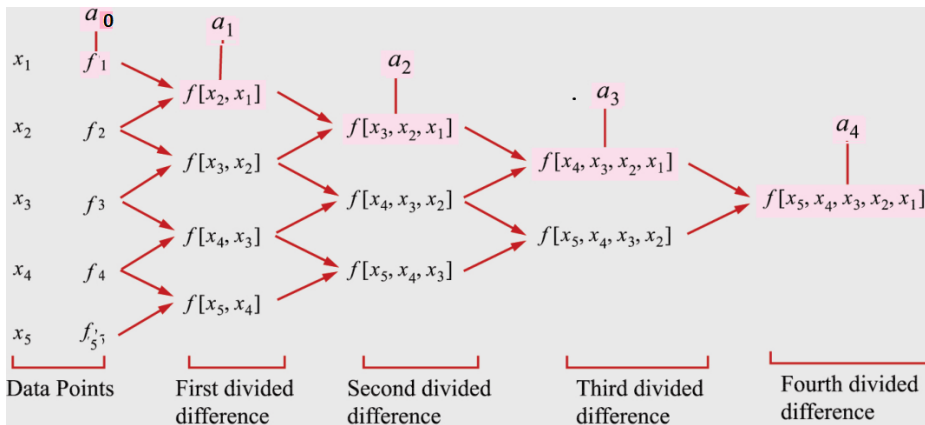
$$f[x_i] = f(x_i) \quad \text{zeroth divided difference}$$

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$$f[x_0, x_1, \cdots, x_k] = \frac{f[x_k, x_{k-1}, \cdots, x_1] - f[x_{k-1}, x_{k-2}, \cdots, x_0]}{x_k - x_0}$$

Newton Divided Difference (NDD) Table



x	$f(x)$	First divided differences	Second divided differences	Third divided differences
x_0	$f[x_0]$			
		$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0}$		
x_1	$f[x_1]$		$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}$	
		$f[x_1, x_2] = \frac{f[x_2] - f[x_1]}{x_2 - x_1}$		$f[x_0, x_1, x_2, x_3] = \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0}$
x_2	$f[x_2]$		$f[x_1, x_2, x_3] = \frac{f[x_2, x_3] - f[x_1, x_2]}{x_3 - x_1}$	
		$f[x_2, x_3] = \frac{f[x_3] - f[x_2]}{x_3 - x_2}$		$f[x_1, x_2, x_3, x_4] = \frac{f[x_2, x_3, x_4] - f[x_1, x_2, x_3]}{x_4 - x_1}$
x_3	$f[x_3]$		$f[x_2, x_3, x_4] = \frac{f[x_3, x_4] - f[x_2, x_3]}{x_4 - x_2}$	
		$f[x_3, x_4] = \frac{f[x_4] - f[x_3]}{x_4 - x_3}$		$f[x_2, x_3, x_4, x_5] = \frac{f[x_3, x_4, x_5] - f[x_2, x_3, x_4]}{x_5 - x_2}$
x_4	$f[x_4]$		$f[x_3, x_4, x_5] = \frac{f[x_4, x_5] - f[x_3, x_4]}{x_5 - x_3}$	
		$f[x_4, x_5] = \frac{f[x_5] - f[x_4]}{x_5 - x_4}$		
x_5	$f[x_5]$			

Newton Divided Difference (NDD) table

For the data

x	1.0	1.3	1.6	1.9	2.2
f(x)	0.7651	0.6201	0.4554	0.2818	0.1103

Newton Divided Difference (NDD) table

For the data

x	1.0	1.3	1.6	1.9	2.2
f(x)	0.7651	0.6201	0.4554	0.2818	0.1103

i	x_i	$f[x_i]$	$f[x_i, x_{i-1}]$	$f[x_i, x_{i-1}, x_{i-2}]$	$f[x_i, x_{i-1}, x_{i-2}, x_{i-3}]$
0	1.0	0.7651			
1	1.3	0.6201			
2	1.6	0.4554			
3	1.9	0.2818			
4	2.2	0.1103			



Newton Divided Difference (NDD) table

For the data

x	1.0	1.3	1.6	1.9	2.2
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i	x_i	$f[x_i]$	$f[x_i, x_{i-1}]$	$f[x_i, x_{i-1}, x_{i-2}]$	$f[x_i, x_{i-1}, x_{i-2}, x_{i-3}]$
0	1.0	0.7651	♡		
1	1.3	0.6201			
2	1.6	0.4554			
3	1.9	0.2818			
4	2.2	0.1103			

♡ The First Divided Difference involving x_0 and x_1 is

$$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0} =$$

Newton Divided Difference (NDD) table

For the data

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f(x)	0.7651	0.6201	0.4554	0.2818	0.1103

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0	1.0	0.7651	♡		
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♡ The First Divided Difference involving x_0 and x_1 is

$$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0} = 0.6201$$

Newton Divided Difference (NDD) table

For the data

x	1.0	1.3	1.6	1.9	2.2
f(x)	0.7651	0.6201	0.4554	0.2818	0.1103

i	x_i	$f[x_i]$	$f[x_i, x_{i-1}]$	$f[x_i, x_{i-1}, x_{i-2}]$	$f[x_i, x_{i-1}, x_{i-2}, x_{i-3}]$
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♡ The First Divided Difference involving x_0 and x_1 is

$$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0} = \frac{0.6201 - 0.7651}{1.3 - 1.0}$$

Newton Divided Difference (NDD) table

For the data

x	1.0	1.3	1.6	1.9	2.2
f(x)	0.7651	0.6201	0.4554	0.2818	0.1103

i	x_i	$f[x_i]$	$f[x_i, x_{i-1}]$	$f[x_i, x_{i-1}, x_{i-2}]$	$f[x_i, x_{i-1}, x_{i-2}, x_{i-3}]$
0	1.0	0.7651	♡		
1	1.3	0.6201			
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♡ The First Divided Difference involving x_0 and x_1 is

$$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0} = \frac{0.6201 - 0.7651}{1.3 - 1.0} =$$

Newton Divided Difference (NDD) table

For the data

x	1.0	1.3	1.6	1.9	2.2
f(x)	0.7651	0.6201	0.4554	0.2818	0.1103

i	x_i	$f[x_i]$	$f[x_i, x_{i-1}]$	$f[x_i, x_{i-1}, x_{i-2}]$	$f[x_i, x_{i-1}, x_{i-2}, x_{i-3}]$
0	1.0	0.7651	♡		
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♡ The First Divided Difference involving x_0 and x_1 is

$$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0} = \frac{0.6201 - 0.7651}{1.3 - 1.0} = -0.4833.$$

Newton Divided Difference (NDD) table

For the data

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f(x)	0.7651	0.6201	0.4554	0.2818	0.1103

i	x_i	$f[x_i]$	$f[x_i, x_{i-1}]$	$f[x_i, x_{i-1}, x_{i-2}]$	$f[x_i, x_{i-1}, x_{i-2}, x_{i-3}]$
0	1.0	0.7651	♡		
1	1.3	0.6201			
2	1.6	0.4554			
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Remaining First Divided Differences are as follows

Newton Divided Difference (NDD) table

For the data

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Newton Divided Difference (NDD) table

For the data

x	1.0	1.3	1.6	1.9	2.2
f(x)	0.7651	0.6201	0.4554	0.2818	0.1103

i	x_i	$f[x_i]$	$f[x_i, x_{i-1}]$	$f[x_i, x_{i-1}, x_{i-2}]$	$f[x_i, x_{i-1}, x_{i-2}, x_{i-3}]$
0	1.0	0.7651			
			-0.4833		
1	1.3	0.6201			
			$f[x_1, x_2]$		
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3	1.9	0.2818			
4	2.2	0.1103			

$f[x_1, x_2]$ involves x_1 and x_2 and given by

$$f[x_1, x_2] = \frac{f[x_2] - f[x_1]}{x_2 - x_1} =$$

Newton Divided Difference (NDD) table

For the data

x	1.0	1.3	1.6	1.9	2.2
f(x)	0.7651	0.6201	0.4554	0.2818	0.1103

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0	1.0	0.7651			
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$$f[x_1, x_2] = \frac{f[x_2] - f[x_1]}{x_2 - x_1} = 0.4554$$

Newton Divided Difference (NDD) table

For the data

x	1.0	1.3	1.6	1.9	2.2
f(x)	0.7651	0.6201	0.4554	0.2818	0.1103

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Newton Divided Difference (NDD) table

For the data

x	1.0	1.3	1.6	1.9	2.2
f(x)	0.7651	0.6201	0.4554	0.2818	0.1103

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0	1.0	0.7651			
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$$f[x_1, x_2] = \frac{f[x_2] - f[x_1]}{x_2 - x_1} = \frac{0.4554 - 0.6201}{1.6 - 1.3} =$$

Newton Divided Difference (NDD) table

For the data

x	1.0	1.3	1.6	1.9	2.2
f(x)	0.7651	0.6201	0.4554	0.2818	0.1103

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$$f[x_1, x_2] = \frac{f[x_2] - f[x_1]}{x_2 - x_1} = \frac{0.4554 - 0.6201}{1.6 - 1.3} = -0.549.$$

Newton Divided Difference (NDD) table

For the data

x	1.0	1.3	1.6	1.9	2.2
f(x)	0.7651	0.6201	0.4554	0.2818	0.1103

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Remaining First Divided Differences are as follows

NDD table for the given data

x	1.0	1.3	1.6	1.9	2.2
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0	1.0	0.7651			
			-0.4833		
1	1.3	0.6201			
			-0.549		
2	1.6	0.4554			
			-0.5787		
3	1.9	0.2818			
			-0.5717		
4	2.2	0.1103			



NDD table for the given data

x	1.0	1.3	1.6	1.9	2.2
f(x)	0.7651	0.6201	0.4554	0.2818	0.1103

i	x_i	$f[x_i]$	$f[x_i, x_{i-1}]$	$f[x_i, x_{i-1}, x_{i-2}]$	$f[x_i, x_{i-1}, x_{i-2}, x_{i-3}]$
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			-0.5717		
4	2.2	0.1103			

♠ The Second Divided Difference involving x_0 , x_1 and x_2 is

$$f[x_2, x_1, x_0] = \frac{f[x_2, x_1] - f[x_1, x_0]}{x_2 - x_0} =$$

NDD table for the given data

x	1.0	1.3	1.6	1.9	2.2
f(x)	0.7651	0.6201	0.4554	0.2818	0.1103

i	x_i	$f[x_i]$	$f[x_i, x_{i-1}]$	$f[x_i, x_{i-1}, x_{i-2}]$	$f[x_i, x_{i-1}, x_{i-2}, x_{i-3}]$
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			-0.5787		
3	1.9	0.2818			
			-0.5717		
4	2.2	0.1103			

♠ The Second Divided Difference involving x_0 , x_1 and x_2 is

$$f[x_2, x_1, x_0] = \frac{f[x_2, x_1] - f[x_1, x_0]}{x_2 - x_0} = \frac{-0.549 - (-0.4833)}{1.6 - 1.0}$$

NDD table for the given data

x	1.0	1.3	1.6	1.9	2.2
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♠ The Second Divided Difference involving x_0 , x_1 and x_2 is

$$f[x_2, x_1, x_0] = \frac{f[x_2, x_1] - f[x_1, x_0]}{x_2 - x_0} = \frac{-0.549 - (-0.4833)}{1.6 - 1.0} = -0.1094$$

NDD table for the given data

x	1.0	1.3	1.6	1.9	2.2
f(x)	0.7651	0.6201	0.4554	0.2818	0.1103

i	x_i	$f[x_i]$	$f[x_i, x_{i-1}]$	$f[x_i, x_{i-1}, x_{i-2}]$	$f[x_i, x_{i-1}, x_{i-2}, x_{i-3}]$
0	1.0	0.7651			
			-0.4833		
1	1.3	0.6201			
			-0.549		
2	1.6	0.4554			
			-0.5787		
3	1.9	0.2818			
			-0.5717		
4	2.2	0.1103			

♠ The Second Divided Difference involving x_0, x_1 and x_2 is

$$f[x_2, x_1, x_0] = \frac{f[x_2, x_1] - f[x_1, x_0]}{x_2 - x_0} = \frac{-0.549 - (-0.4833)}{1.6 - 1.0} = -0.1094$$

Remaining Second Divided Difference are as follows

x	1.0	1.3	1.6	1.9	2.2
f(x)	0.7651	0.6201	0.4554	0.2818	0.1103

i	x_i	$f[x_i]$	$f[x_i, x_{i-1}]$	$f[x_i, x_{i-1}, x_{i-2}]$	$f[x_i, x_{i-1}, x_{i-2}, x_{i-3}]$
0	1.0	0.7651			
			-0.4833		
1	1.3	0.6201		-0.1094	
			-0.549		
2	1.6	0.4554		-0.0494	
			-0.5787		
3	1.9	0.2818		0.0117	
			-0.5717		
4	2.2	0.1103			



x	1.0	1.3	1.6	1.9	2.2
f(x)	0.7651	0.6201	0.4554	0.2818	0.1103

i	x_i	$f[x_i]$	$f[x_i, x_{i-1}]$	$f[x_i, x_{i-1}, x_{i-2}]$	$f[x_i, x_{i-1}, x_{i-2}, x_{i-3}]$
0	1.0	0.7651			
			-0.4833		
1	1.3	0.6201		-0.1094	
			-0.549		♣
2	1.6	0.4554		-0.0494	
			-0.5787		
3	1.9	0.2818		0.0117	
			-0.5717		
4	2.2	0.1103			

♣ The Third Divided Difference $f[x_3, x_2, x_1, x_0]$

$$= \frac{f[x_3, x_2, x_1] - f[x_2, x_1, x_0]}{x_3 - x_0} = \frac{-0.1094 - (-0.494)}{1.9 - 1.0} = -0.0667$$

x	1.0	1.3	1.6	1.9	2.2
f(x)	0.7651	0.6201	0.4554	0.2818	0.1103

i	x_i	$f[x_i]$	$f[x_i, x_{i-1}]$	$f[x_i, x_{i-1}, x_{i-2}]$	$f[x_i, x_{i-1}, x_{i-2}, x_{i-3}]$
0	1.0	0.7651			
			-0.4833		
1	1.3	0.6201		-0.1094	
			-0.549		♣
2	1.6	0.4554		-0.0494	
			-0.5787		
3	1.9	0.2818		0.0117	
			-0.5717		
4	2.2	0.1103			

♣ The Third Divided Difference $f[x_3, x_2, x_1, x_0]$

$$= \frac{f[x_3, x_2, x_1] - f[x_2, x_1, x_0]}{x_3 - x_0} = \frac{-0.1094 - (-0.494)}{1.9 - 1.0} = -0.0667$$

Remaining Third Divided Differences are as follows

x	1.0	1.3	1.6	1.9	2.2
f(x)	0.7651	0.6201	0.4554	0.2818	0.1103

i	x_i	$f[x_i]$	$f[x_i, x_{i-1}]$	$f[x_i, x_{i-1}, x_{i-2}]$	$f[x_i, x_{i-1}, x_{i-2}, x_{i-3}]$
0	1.0	0.7651			
			-0.4833		
1	1.3	0.6201		-0.1094	
			-0.549		-0.0667
2	1.6	0.4554		-0.0494	
			-0.5787		0.0679
3	1.9	0.2818		0.0117	
			-0.5717		
4	2.2	0.1103			

x	1.0	1.3	1.6	1.9	2.2
f(x)	0.7651	0.6201	0.4554	0.2818	0.1103

i	x_i	$f[x_i]$	$f[x_i, x_{i-1}]$	$f[x_i, x_{i-1}, x_{i-2}]$	$f[x_i, x_{i-1}, x_{i-2}, x_{i-3}]$
0	1.0	0.7651			
			-0.4833		
1	1.3	0.6201		-0.1094	
			-0.549		-0.0667
2	1.6	0.4554		-0.0494	
			-0.5787		0.0679
3	1.9	0.2818		0.0117	
			-0.5717		
4	2.2	0.1103			

The Fourth Divided Difference $f[x_4, x_3, x_2, x_1, x_0]$

$$= \frac{f[x_4, x_3, x_2, x_1] - f[x_3, x_2, x_1, x_0]}{x_4 - x_0} = \frac{-0.0679 - (-0.0667)}{2.2 - 1.0} = -0.001$$

x	1.0	1.3	1.6	1.9	2.2
f(x)	0.7651	0.6201	0.4554	0.2818	0.1103

i	x_i	$f[x_i]$	$f[x_i, x_{i-1}]$	$f[x_i, x_{i-1}, x_{i-2}]$	$f[x_i, x_{i-1}, x_{i-2}, x_{i-3}]$
0	1.0	0.7651			
			-0.4833		
1	1.3	0.6201		-0.1094	
			-0.549		-0.0667
2	1.6	0.4554		-0.0494	
			-0.5787		0.0679
3	1.9	0.2818		0.0117	
			-0.5717		
4	2.2	0.1103			

The Fourth Divided Difference $f[x_4, x_3, x_2, x_1, x_0]$

$$= \frac{f[x_4, x_3, x_2, x_1] - f[x_3, x_2, x_1, x_0]}{x_4 - x_0} = \frac{-0.0679 - (-0.0667)}{2.2 - 1.0} = -0.001$$

This would be last divided difference in this case.

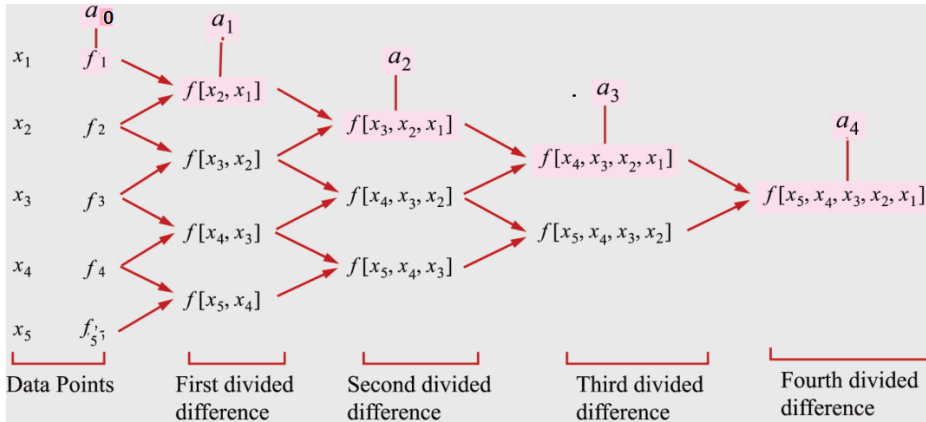
The Newton forward divided-difference polynomial is

$$\begin{aligned}
 P_4(x) = & \textcolor{red}{0.7651} - \textcolor{blue}{0.4833}(x - 1) - \textcolor{green}{0.1094}(x - 1)(x - 1.3) + \\
 & - \textcolor{orange}{0.0667}(x - 1)(x - 1.3)(x - 1.6) \\
 & - \textcolor{violet}{0.001}(x - 1)(x - 1.3)(x - 1.6)(x - 1.9)
 \end{aligned}$$

i	x_i	$f[x_i]$	$f[x_i, x_{i-1}]$	$f[x_i, x_{i-1}, x_{i-2}]$	$f[x_i, x_{i-1}, x_{i-2}, x_{i-3}]$
0	1.0	$\textcolor{red}{0.7651}$			
1	1.3	0.6201	$\textcolor{blue}{-0.4833}$		
2	1.6	0.4554	-0.549	$\textcolor{green}{-0.1094}$	
3	1.9	0.2818	-0.5787	-0.0494	$\textcolor{orange}{-0.0667}$
4	2.2	0.1103	-0.5717	0.0117	0.0679

$$f[x_4, x_3, x_2, x_1, x_0] = \textcolor{violet}{-0.001}$$

$$P(x) = \mathbf{a}_0 + \mathbf{a}_1(x - x_0) + \mathbf{a}_2(x - x_0)(x - x_1) + \cdots + \mathbf{a}_n(x - x_0) \cdots (x - x_{n-1})$$



To Study in Next Video

- Newton's Divided Difference Interpolation solved example
- Forward/Backward Difference Interpolation
- Equally Spaced Data